Uniform boundary probability for multiplicatively weighted random integer factorizations

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doi:10.5802/jep.317; CC BY4.0

Editorial note. The counted item is the number-theoretic boundary-event lemma, not the separate PD-radius lemma or the old coupling theorem.

Editorial note. The constants sought are uniform in alpha_i,u,k and the permitted allocation law f; the two complementary hyperbola regions are essential for retaining that scope.

Editorial note. The first mean-value estimate1682 is a named standard prerequisite. Its local weighted harmonic-window consequence1683 and proof1687-1700 remain in the solution chain.

Editorial note. At source1699 the numerical intermediate count of the logarithmic partition intervals is understated: the defining interval1696 has logarithmic length1+2log y. The proof uses only O(log y), which is also stated on1699 and is valid uniformly fory>=2. The original author line remains unchanged; the implicit absolute constant absorbs this numerical count.

Editorial scope. One integer-factorization boundary bound. Its square-full moment and weighted harmonic-window auxiliaries are not separate counted items. The full Dirichlet-law conclusion is not counted on top of these prepared estimates.

Allowed standard prerequisites (editorial).

- Mean-value upper bound for alpha to the number of distinct prime factors. Uniformly for alpha in[0,1] and y>=2, sum_(m<=y) alpha^(omega(m)) <= C y(log y)^(alpha-1).. Bound one factor after the proof-specific square-free decomposition and hyperbola split. The short harmonic window bound is proved locally.
- Classical Mertens theorem and partial summation. Prove the local weighted harmonic-window estimate1678-1700.
- Counting square-full integers. There are O(sqrt z) square-full integers at most z; each can be written a^2b^3 with b square-free, and sum_b b^(-3/2) converges.. Bound the square-full allocation count and the moment sum. No boundary probability estimate is supplied.

Source reference. Original Theorem 2: sources/arratia-dirichlet-jep-2025-317/source0.tex, source lines 349–359. The definition footnote says the allocation conditions will

be used for this later theorem; that theorem is not used to prove the present integer boundary estimate.

Let N_x be an integer chosen uniformly at random from the set $\mathbb{Z} \cap [1, x]$. We may then factor it uniquely as $N_x = P_1 P_2 \cdots$ with the P_i's forming a non-increasing sequence of primes or ones. In 1972, Billingsley [5] showed that, for any fixed positive integer r, the joint distribution of the random vector

Let Δ^{k-1} be the set of k-tuples $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_k) \in \mathbb{R}^k_{\geq 0}$ satisfying $\alpha_1 + \cdots + \alpha_k = 1$. We also need to define a special class of functions:

Definition 0.1 (The class of functions $\mathcal{F}_k(\boldsymbol{\alpha})$)**.** Given $k \in \mathbb{Z}_{\geq 2}$ and $\boldsymbol{\alpha} \in \Delta^{k-1}$, let $\mathcal{F}_k(\boldsymbol{\alpha})$ be the set of functions $f \colon \mathbb{N}^k \to \mathbb{R}_{\geq 0}$ satisfying the following three properties:

a. We have $\sum_{d_1 \cdots d_k = n} f(d_1, \ldots, d_k) = 1$ for all $n \in \mathbb{N}$. Thus, for each fixed $n \in \mathbb{N}$, the function $f(\boldsymbol{d})$ defines a probability mass function on the set of vectors $\boldsymbol{d} \in \mathbb{N}^k$ satisfying $d_1 \cdots d_k = n$.

b. Whenever $\boldsymbol{d}$ satisfies
$$d_i \coloneqq \begin{cases} p & \text{if } i = j, \\ 1 & \text{if } i \neq j. \end{cases}$$
for some $1 \leq j \leq k$ and prime p, then $f(\boldsymbol{d}) = \alpha_j$.

c. The function f is *multiplicative*, i.e. for any vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{N}^k$ with $a_1 \cdots a_k$ and $b_1 \cdots b_k$ being co-prime, we have the property
$$f(a_1 b_1, \ldots, a_k b_k) = f(a_1, \ldots, a_k) \cdot f(b_1, \ldots, b_k).$$

Remark. Let $\omega(n)$ denote the number of distinct prime factors of n. If $d_1 \cdots d_k$ is a square-free number and $f \in \mathcal{F}_k(\boldsymbol{\alpha})$, then properties (b) and (c) of the definition above imply that
$$f(d_1, \ldots, d_k) = \prod_{j=1}^{k} \alpha_j^{\omega(d_j)}. \tag{1}$$

We will use the class of functions $\mathcal{F}_k(\boldsymbol{\alpha})$ to define certain "random factorizations" into k parts of a random integer. Specifically, let us fix $f \in \mathcal{F}_k(\boldsymbol{\alpha})$ and $x \geq 1$. We then define a *random k-factorization corresponding to f* to be a random vector $\boldsymbol{D}_{f,x} = (D_{f,x,1}, \ldots, D_{f,x,k})$ taking values on $\mathbb{N}^k$ and satisfying the formula[1]
$$\mathbb{P}\Big[D_{f,x,i} = d_i \ \ \forall i \leq k \ \Big| \ N_x = n \Big] = f(d_1, \ldots, d_k) \tag{2}$$
for all $n \in \mathbb{N}$ and all k-tuples $(d_1, \ldots, d_k) \in \mathbb{N}^k$ with $d_1 \cdots d_k = n$.

[1]Strictly speaking, we only need property (a) of Definition 0.1 to define $\boldsymbol{D}_{f,x}$. But we will also need the other two properties when proving Theorem 2 below.

0.1 Notation

We let $\log_j$ denote the j-iteration of the natural logarithm, meaning that $\log_1 = \log$ and $\log_j = \log \circ \log_{j-1}$ for $j \geq 2$.

Throughout the paper, the letter p is reserved for prime numbers and the letter n is reserved for natural numbers, unless stated otherwise. Given such p and n, we write $\nu_p(n)$ for the p-adic valuation of n, that is to say the largest integer $v \geq 0$ such that $p^v | n$. In addition, we write $\omega(n)$ for the number of distinct prime factors of n.

Moreover, for $n \in \mathbb{N}$, we let $s(n)$ denote its largest square-full divisor. Also, we let $n^\flat := n/s(n)$ and we note that $n^\flat$ is always square-free and co-prime to $s(n)$.

We write $\pi(x)$ for the number of primes $\leq x$. We shall also use heavily Chebyshev's function $\theta(x) := \sum_{p \leq x} \log p$.

To describe various estimates, we use Vinogradov's notation $f(x) \ll g(x)$ or Landau's notation $f(x) = O(g(x))$ to mean that $|f(x)| \leq C \cdot g(x)$ for a positive constant C. If C depends on a parameter α, we write $f(x) \ll_\alpha g(x)$ or $f(x) = O_\alpha(g(x))$. If two positive functions f, g have the same order of magnitude in the sense that $f(x) \ll g(x) \ll f(x)$, then we write $f(x) \asymp g(x)$.

If P is some proposition, then the indicator function $\mathbb{1}_P$ will be equal to 1 if P is true and 0 if P is false.

$$\mathcal{E}_{\mathrm{NT}}(\Delta_{\mathbf{u}}^{k-1})^c \subseteq \bigcup_{\substack{i<k \\ u_i \neq 1}} \mathcal{B}_{\mathrm{NT}}^{(i)}(u_i), \quad \text{where} \quad \mathcal{B}_{\mathrm{NT}}^{(i)}(u) := \Big\{ \big| \log(D_{f,x,i}/x^u) \big| \leq 3 \log(x/N_x^\flat) \Big\},$$

Proposition 0.2 (The prime number theorem)**.** *For $x \geq 2$, we have*

$$\theta(x) = x + O(x/(\log x)^3)$$

and

$$\psi(x) = x + O(x/(\log x)^3).$$

We shall also need a stronger version of Mertens's estimate, which in its classical form, yields an estimate for the product $\prod_{p \leq x}(1 - 1/p)$.

Proposition 0.3 (Strong Mertens's estimate)**.** *For $x \geq 2$, we have*

$$\sum_{p^k \leq x} \frac{1}{kp^k} = \log_2 x + \gamma + O(1/(\log x)^3).$$

Proof. Using the prime number theorem (Proposition 0.2), there exists a constant c_0 satisfying

$$\sum_{p^k \leq x} \frac{1}{kp^k} = \int_{2^-}^{x} \frac{\mathrm{d}\psi(t)}{t \log t} = \log_2 x + c_0 + O(1/(\log x)^3).$$

for $x \geq 2$. For the proof of $\gamma = c_0$, see the proof of [17, Th. 2.7]. □

Lemma 0.4. *For $i \in \{1, 2, \ldots, k-1\}$ and $u \in (0, 1)$, we have the uniform estimate*

$$\mathbb{P}\big[\mathcal{B}_{\mathrm{NT}}^{(i)}(u)\big] \ll \frac{1}{(1 + u \log x)^{1-\alpha_i} (1 + (1-u) \log x)^{\alpha_i}}.$$

Lemma 0.5 (Properties of N_x)**.** *Fix $\alpha \in [0,1)$ and $\beta \in [0,1/2)$. Uniformly over $x \geq 1$, we have*

$$\mathbb{E}\big[(x/N_x)^\alpha s(N_x)^\beta\big] \ll_{\alpha,\beta} 1.$$

Proof. We must show that

$$S \coloneqq \sum_{n\leq x} (x/n)^\alpha s(n)^\beta \ll_{\alpha,\beta} x.$$

Indeed, if we let $b = s(n)$ and $a = n^\flat = n/b$, then

$$S \leq \sum_{\substack{b\leq x\\ b \text{ square-full}}} x^\alpha b^{\beta-\alpha} \sum_{a\leq x/b} a^{-\alpha} \ll_\alpha \sum_{\substack{b\leq x\\ b \text{ square-full}}} x^\alpha b^{\beta-\alpha}(x/b)^{1-\alpha} \leq x \sum_{b \text{ square-full}} b^{\beta-1},$$

where we used our assumption that $\alpha < 1$. Since we have assumed that $\beta < 1/2$, the last sum over b converges, thus completing the proof. □

1 Proof of Lemma 0.4 using elementary number-theoretic techniques

Note that

$$\mathbb{P}\big[N_x^\flat \leq x/(\log x)^3\big] \leq \frac{1}{\log x}\mathbb{E}\big[(x/N_x^\flat)^{1/3}\big] \ll \frac{1}{\log x} \tag{3}$$

by Lemma 0.5 applied with $\alpha = \beta = 1/3$.

Fix now $i \in \{1,\dots,k-1\}$ and $u \in (0,1)$. For simplicity, let us write

$$\alpha \coloneqq \alpha_i$$

for the remainder of this section. Given z, let us define

$$S(z) \coloneqq \sum_{\substack{n\leq x\\ n^\flat\in(x/z,2x/z]}} \sum_{\substack{d_1\cdots d_k=n\\ d_i\in[x^u z^{-3},x^u z^3]}} f(d_1,\dots,d_k).$$

Using (3) and (2), we find that

$$\mathbb{P}\big[\mathcal{B}_{\mathrm{NT}}^{(i)}(u)\big] \leq \frac{1}{\lfloor x\rfloor} \sum_{1\leq m\leq 5\log_2 x} S(2^m) + O\big(1/\log x\big). \tag{4}$$

Hence Lemma 0.4 follows readily from the estimate below:

Lemma 1.1. *There exists a universal constant $x_0 \geq 2$ such that if $x \geq x_0$ and $z \in [2,(\log x)^4]$, then we have the uniform estimate*

$$S(z) \ll \frac{\log z}{\sqrt{z}} \cdot \frac{x}{(1+u\log x)^{1-\alpha}(1+(1-u)\log x)^\alpha}.$$

Proof. If $d_1\cdots d_k = n$, then we may uniquely write $d_j = d_j' d_j''$ with $d_j' | n^\flat$ and $d_j'' | s(n)$. In particular, $d_j'' \leq s(n) \leq x/n^\flat \leq z$ for all j. Hence, if

$d_i \in [x^u z^{-3}, x^u z^3]$, we must also have that $d_i' \in [x^u z^{-4}, x^u z^3]$. Hence, using property (c) of Definition 0.1, we deduce that

$$\sum_{\substack{d_1\cdots d_k=n\\ d_i\in[x^u z^{-3},x^u z^3]}} f(d_1,\dots,d_k) \le \sum_{\substack{d_1'\cdots d_k'=n^\flat\\ d_i'\in[x^u z^{-4},x^u z^3]}} f(d_1',\dots,d_k') \sum_{d_1''\cdots d_k''=s(n)} f(d_1'',\dots,d_k'').$$

Relation (1) implies that $f(d_1',\dots,d_k') = \prod_{j=1}^k \alpha_j^{\omega(d_j')}$. Moreover, using property (a) of Definition 0.1, we find that the sum over the d_j'''s equals 1. In conclusion,

$$\begin{aligned}\sum_{\substack{d_1\cdots d_k=n\\ d_i\in[x^u z^{-3},x^u z^3]}} f(d_1,\dots,d_k) &\le \sum_{\substack{d_1'\cdots d_k'=n^\flat\\ d_i'\in[x^u z^{-4},x^u z^3]}} \prod_{j=1}^k \alpha_j^{\omega(d_j')}\\ &= \sum_{\substack{d\mid n^\flat\\ d\in[x^u z^{-4},x^u z^3]}} \alpha^{\omega(d)}(1-\alpha)^{\omega(n^\flat/d)}.\end{aligned}$$

Let $n^\flat = dm$ and $s(n) = b$. If $n^\flat \in [x/z, 2x/z]$ and $d \in [x^u z^{-4}, x^u z^3]$, then $m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]$. Hence, we find that

$$\begin{aligned}S(z) &\le \sum_{\substack{dm\in[x/z,2x/z]\\ d\in[x^u z^{-4},x^u z^3]\\ m\in[x^{1-u} z^{-4},2x^{1-u} z^3]}}\sum \alpha^{\omega(d)}(1-\alpha)^{\omega(m)} \sum_{\substack{b\le z\\ b \text{ square-full}}} 1\\ &\ll \sqrt{z} \sum_{\substack{dm\in[x/z,2x/z]\\ d\in[x^u z^{-4},x^u z^3]\\ m\in[x^{1-u} z^{-4},2x^{1-u} z^3]}}\sum \alpha^{\omega(d)}(1-\alpha)^{\omega(m)} = \sqrt{z}\cdot(S_1+S_2),\end{aligned} \tag{5}$$

where S_1 denotes the double sum over d and m with the additional constraint $d \le \sqrt{2x/z}$, and S_2 is the corresponding sum over pairs (d,m) with $d > \sqrt{2x/z}$ and $m \le \sqrt{2x/z}$.

First, we bound S_1. Note that for the conditions on d to be compatible we must have that $u \le 2/3$, provided that x is large enough. Assuming this is the case, we apply twice Proposition 1.2 to find that

$$\begin{aligned}S_1 &\le \sum_{\substack{d\le\sqrt{2x/z}\\ d\in[x^u z^{-4},x^u z^3]}} \alpha^{\omega(d)} \sum_{m\le 2x/dz} (1-\alpha)^{\omega(m)}\\ &\ll \sum_{\substack{d\le\sqrt{2x/z}\\ d\in[x^u z^{-4},x^u z^3]}} \alpha^{\omega(d)}\cdot \frac{x(\log x)^{-\alpha}}{dz}\\ &\ll \frac{x(\log z)(1+u\log x)^{\alpha-1}(\log x)^{-\alpha}}{z}.\end{aligned} \tag{6}$$

We bound S_2 in a similar manner. For this sum to be non-empty, we have

$u \geq 1/3$. If this is the case, then we have

$$\begin{aligned} S_2 &\leq \sum_{\substack{m \leq \sqrt{2x/z} \\ m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]}} (1-\alpha)^{\omega(m)} \sum_{d \leq 2x/mz} \alpha^{\omega(d)} \\ &\ll \sum_{\substack{m \leq \sqrt{2x/z} \\ m \in [x^{1-u} z^{-4}, 2x^{1-u} z^3]}} (1-\alpha)^{\omega(m)} \cdot \frac{x(\log x)^{\alpha-1}}{mz} \qquad (7) \\ &\ll \frac{x(\log z)(1+(1-u)\log x)^{-\alpha}(\log x)^{\alpha-1}}{z}. \end{aligned}$$

Combining (5), (6) and (7), while keeping in mind that $S_1 = 0$ if $u > 2/3$ and that $S_2 = 0$ if $u < 1/3$, completes the proof of the lemma, □

Proposition 1.2. *Uniformly for $\alpha \in [0,1]$ and $x, y \geq 2$, we have*

$$\sum_{n \leq x} \alpha^{\omega(n)} \ll x(\log x)^{\alpha-1} \quad \text{and} \quad \sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \ll (\log y)(\log(xy))^{\alpha-1}.$$

Proof. The first bound follows readily with [13, Th. 14.2]. Let us now prove the second one.

When $y \geq \sqrt{x}$, we have

$$\sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \leq \sum_{p \mid n \Rightarrow p \leq y^3} \frac{\alpha^{\omega(n)}}{n} = \prod_{p \leq y^3} \Big(1 + \frac{\alpha}{p-1}\Big) \ll (\log y)^{\alpha}$$

by the inequality $1 + t \leq e^t$ and Mertens's theorem (Proposition 0.3). This completes the proof in this case. Finally, assume that $y \leq \sqrt{x}$. We then have

$$\sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \leq \sum_{\substack{m \in \mathbb{Z} \\ e^m \in [x/y, exy]}} \sum_{n \in (e^{m-1}, e^m]} \frac{\alpha^{\omega(n)}}{n} \leq \sum_{\substack{m \in \mathbb{Z} \\ e^m \in [x/y, exy]}} e^{1-m} \sum_{n \leq e^m} \alpha^{\omega(n)}.$$

For every m as above, the innermost sum is $\ll e^m m^{\alpha-1} \asymp e^m (\log(xy))^{\alpha-1}$ by the first part of the lemma and our assumption that $y \leq \sqrt{x}$. Since there are $\leq 1 + \log y \ll \log y$ choices for m, the needed estimate follows in this last case too. □

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